

Problem Set 1

CMSC 27200: Theory of Algorithms

Problem 1: Gale-Shapley and Stable Matchings (25 points, 5 extra credit points available)

- (a) 15 points: Given the following preference lists, run the Gale-Shapley algorithm with group A making the offers to obtain a stable matching.

Group A 's preference lists (from most preferred to least preferred):

a_1 : b_3, b_2, b_1, b_4

a_2 : b_1, b_4, b_3, b_2

a_3 : b_4, b_1, b_3, b_2

a_4 : b_1, b_4, b_2, b_3

Group B 's preference lists (from most preferred to least preferred):

b_1 : a_1, a_3, a_4, a_2

b_2 : a_3, a_4, a_2, a_1

b_3 : a_3, a_2, a_1, a_4

b_4 : a_1, a_4, a_3, a_2

For your answer (and for your answer to part b as well), you should give the following data. At each step, give the offer that is made, whether or not this offer is accepted, and the resulting partial matching.

Solution:

1. a_1 makes offer to b_3 : accepted as b_3 is unmatched
Current partial matching: (a_1, b_3)
2. a_2 makes offer to b_1 : accepted as b_1 is unmatched
Current partial matching: $(a_1, b_3), (a_2, b_1)$
3. a_3 makes offer to b_4 : accepted as b_4 is unmatched
Current partial matching: $(a_1, b_3), (a_2, b_1), (a_3, b_4)$
4. a_4 makes offer to b_1 : accepted as b_1 prefers a_4 to a_2
Current partial matching: $(a_1, b_3), (a_3, b_4), (a_4, b_1)$

5. a_2 makes offer to b_4 : rejected as b_4 prefers a_3 to a_2
Current partial matching: $(a_1, b_3), (a_3, b_4), (a_4, b_1)$
6. a_2 makes offer to b_3 : accepted as b_3 prefers a_2 to a_1
Current partial matching: $(a_3, b_4), (a_4, b_1), (a_2, b_3)$
7. a_1 makes offer to b_2 : accepted as b_2 is unmatched
Final matching: $(a_3, b_4), (a_4, b_1), (a_2, b_3), (a_1, b_2)$

- (b) 10 points: Now run the Gale-Shapley algorithm with group B making the offers to obtain another stable matching. Which people are happier in this new stable matching (compared to the stable matching found in part a)?

Solution:

1. b_1 makes offer to a_1 : accepted as a_1 is unmatched
Current partial matching: (a_1, b_1)
2. b_2 makes offer to a_3 : accepted as a_3 is unmatched
Current partial matching: $(a_1, b_1), (a_2, b_3)$
3. b_3 makes offer to a_3 : accepted as a_3 prefers b_3 to b_2
Current partial matching: $(a_1, b_1), (a_3, b_3)$
4. b_4 makes offer to a_1 : rejected as a_1 prefers b_1 to b_4
Current partial matching: $(a_1, b_1), (a_3, b_3)$
5. b_4 makes offer to a_4 : accepted as a_4 is unmatched
Current partial matching: $(a_1, b_1), (a_3, b_3), (a_4, b_4)$
6. b_2 makes offer to a_4 : rejected as a_4 prefers b_4 to b_2
Current partial matching: $(a_1, b_1), (a_3, b_3), (a_4, b_4)$
7. b_2 makes offer to a_2 : accepted as a_2 is unmatched
Final matching: $(a_1, b_1), (a_3, b_3), (a_4, b_4), (a_2, b_2)$

b_1, b_2, b_3 , and b_4 are happier in the new stable matching as they are each matched with someone higher on their preference list.

- (c) 5 points extra credit: What other stable matching(s) are there, if any? Note: For full extra credit you should show that you have indeed found all of the possible stable matchings.

Hint is available.

Solution: Let S_i denote the set of possible matches for each a_i . We use the fact that when group A makes offers, G-S gives the best possible matching for A , and when group B makes offers, G-S gives the worst possible matching for A (Theorem 1.9). Because the best and worst possible matchings are consecutive on a_1, a_2 , and a_4 's preference lists, we $S_1 = \{b_2, b_1\}$, $S_2 = \{b_3, b_2\}$, and $S_4 = \{b_1, b_4\}$. For a_3 , we know $\{b_4, b_3\} \subseteq S_3$, but we don't know yet if $b_1 \in S_3$.

If we match a_3 with b_4 , then by elimination, we must match a_4 with b_1 , a_2 with b_3 , and a_1

with b_2 (giving the G-S matching when group A offers). Similarly, if we match a_3 with b_3 , we get the G-S matching when group B offers. Finally, if we match a_3 with b_1 , we must match a_1 with b_2 , a_2 with b_3 , and a_4 with b_4 . To see that $M = \{(a_1, b_2), (a_2, b_3), (a_3, b_1), (a_4, b_4)\}$ is a stable matching, we make the following observations:

1. The only person who a_1 prefers to b_2 is b_3 , but b_3 prefers a_2 to a_1 . Thus, there are no unstable pairs involving a_1 .
2. While a_2 prefers b_1 and b_4 to b_3 , b_1 prefers a_3 to a_2 and b_4 prefers a_4 to a_2 . Thus, there are no unstable pairs involving a_2 .
3. The only person who a_3 prefers to b_1 is b_4 , but b_4 prefers a_4 to a_3 . Thus, there are no unstable pairs involving a_3 .
4. The only person who a_4 prefers to b_4 is b_1 , but b_1 prefers a_3 to a_4 . Thus, there are no unstable pairs involving a_4 .

There were maximum 3 possibilities for a_3 's partner, and each resulted in a unique stable matching, so there are exactly 3 stable matchings. In conclusion, the only other stable matching is $\{(a_1, b_2), (a_2, b_3), (a_3, b_1), (a_4, b_4)\}$.

Problem 2: Big O Notation (25 points)

- (a) 15 points: Given each of the following statements, what can we say about $T(n)$ using Big O notation? You are asked to give the *best possible answers* using the concepts of $O(\cdot)$, $\Omega(\cdot)$, and $\Theta(\cdot)$. If the given statement is not sufficient to decide that $T(n)$ is $O(\cdot)$, $\Omega(\cdot)$, or $\Theta(\cdot)$ of another simple function, you should indicate this in your answer. All logarithms are base 2.

1. $T(n) \leq 3n + 2n^2 - 15$.

Solution: $O(n^2)$ as $2n^2$ is the fastest-growing term. There is no lower bound given, so we cannot conclude $\Omega(\cdot)$ or $\Theta(\cdot)$.

2. $T(n) = \frac{\sqrt{n}}{2} + 7(\log n)^{10}$.

Solution: $\Theta(\sqrt{n})$ because $O(\sqrt{n})$ and $\Omega(\sqrt{n})$. To compare $\frac{\sqrt{n}}{2}$ and $7 \log^{10} n$, we ignore constants and compare the log of each: $\log(\sqrt{n}) = \log(n^{1/2}) = \frac{1}{2} \log n$ and $\log(\log^{10} n) = 10 \log(\log n)$. Since $\log n$ grows faster than $\log(\log n)$, $\frac{\sqrt{n}}{2}$ is the fastest-growing term.

3. $T(n) \geq 10n^4 + 3n(1.1)^n$.

Solution: $\Omega(n(1.1)^n)$ because $3n(1.1)^n$, as an exponential, is the fastest growing term. There is no upper bound given, so we cannot conclude $O(\cdot)$ or $\Theta(\cdot)$.

4. $2^n \leq T(n) \leq 7n^2 2^{2n}$.

Solution: $O(n^2 2^{2n})$ and $\Omega(2^n)$. Since the $O(\cdot)$ and $\Omega(\cdot)$ don't match, we cannot conclude $\Theta(\cdot)$.

5.
$$T(n) = \begin{cases} 14 - 5n + \frac{n^2}{2} & \text{if } n \bmod 2 = 0 \\ 3n \log n + 100n & \text{if } n \bmod 2 = 1 \end{cases}$$

Solution: $O(n^2)$ and $\Omega(n \log n)$. The upper bound ($O(n^2)$) is determined by the

fastest-growing case (even n), while the lower bound ($\Omega(n \log n)$) is determined by the slowest-growing case (odd n). Since the $O(\cdot)$ and $\Omega(\cdot)$ don't match, we cannot conclude $\Theta(\cdot)$ for $T(n)$ overall.

- (b) 10 points: Order the functions below in order of increasing asymptotic growth. If two functions have the same asymptotic growth, you should indicate this. All logarithms are base 2.
 2^n , n , $\log n$, n^2 , $n \log n$, $n^{\log n}$, $2^{2 \log n}$, 2^{2n} , $n^2 / \log n$, $n 2^n$, $2^{\log \log n}$, $2^{\sqrt{\log n}}$

Hint is available.

Solution: $\log n = 2^{\log \log n} < 2^{\sqrt{\log n}} < n < n \log n < n^2 / \log n < 2^{2 \log n} = n^2 < n^{\log n} < 2^n < n 2^n < 2^{2n}$.

Problem 3: Maximum Unhappiness for Gale-Shapley (20 points, 5 extra credit points available)

Let integer $n \geq 2$. Let's say that we run the Gale-Shapley algorithm on groups $A = \{a_1, \dots, a_n\}$ and $B = \{b_1, \dots, b_n\}$ with group A making the offers.

- (a) 10 points: Is it possible that everyone in group B is matched with their least preferred partner? Either give preference lists where this occurs (your construction should be in terms of n) or prove that this is impossible.

Solution: Yes. For each $i \in [n]$, let a_i have b_i as their first choice and let b_i have a_i as their last choice. Hence, each a_i will make an offer to b_i , who will accept because they are available, and the algorithm will terminate with no rejections.

- (b) 10 points: Is it possible that everyone in group A is matched with their least preferred partner? Either give preference lists where this occurs (your construction should be in terms of n) or prove that this is impossible.

Hint is available.

Solution: This is impossible for $n \geq 2$. Consider the first time any member of group A , denoted a_i , makes an offer to their least preferred partner, denoted b_j . At this point, every other member of group B must have been unavailable to a_i . As unavailable partners never become available again (Lemma 1.6), every other member of group B is currently matched so there are at least $n - 1$ matchings. As a_i is unmatched, there are at most $n - 1$ matchings. Hence, there are exactly $n - 1$ matchings and every member of group B except b_j is matched, so b_j is available and will accept a_i 's offer. The algorithm will then terminate. Therefore, at most one member in group A can be matched with their least preferred partner.

- (c) 5 points extra credit: What is the maximum total number of offers that group A has to make before a stable matching is reached (in terms of n)? For full extra credit, you should give preference lists where this maximum is obtained and explain why this maximum is obtained.

Solution: In part (b), we proved that at most 1 member of group A can be matched with their least preferred partner. To maximize the number of offers, we ask if the next "worst-case" scenario is possible: one member of group A being matched to their least preferred partner,

and every other member being matched to their second-least preferred partner. This would result in $(n-1)(n-1) + n = n^2 - n + 1$ offers.

One set of preference lists for which this occurs is as follows:

$a_1: b_1, b_2, \dots, b_{n-2}, b_{n-1}, b_n$
 $a_2: b_2, b_3, \dots, b_{n-1}, b_1, b_n$
 $\dots$
 $a_{n-1}: b_{n-1}, b_1, \dots, b_{n-3}, b_{n-2}, b_n$
 $a_n: b_1, b_2, \dots, b_{n-2}, b_{n-1}, b_n$

 $b_1: a_2, a_3, \dots, a_{n-1}, a_n, a_1$
 $b_2: a_3, a_4, \dots, a_n, a_1, a_2$
 $\dots$
 $b_{n-1}: a_n, a_1, \dots, a_{n-3}, a_{n-2}, a_{n-1}$
 $b_n: a_1, a_2, \dots, a_{n-2}, a_{n-1}, a_n$ (actually, any preference list for b_n would work)

If we run G-S with group A making the offers, we can have the people in group A make offers in clockwise order $(a_1, \dots, a_n)$. This goes through $n-1$ cycles. For all $i \in [n]$ and $j \in [n-1]$, a_i makes its j th offer to $b_{(i+j-1) \bmod (n-1)}$ and this offer is accepted, displacing $a_{(i+1) \bmod n}$ if $j > 1$ or $i = n$.

After everyone in group A has made $n-1$ offers, a_1 is unmatched and makes an offer to b_n which gives the final matching $M = \{(a_1, b_n), (a_2, b_1), \dots, (a_n, b_{n-1})\}$.

Problem 4: Lying Gale-Shapley? (20 points)

Give an example of the Stable Matching Problem for $n = 3$, i.e., the groups are $A = \{a_1, a_2, a_3\}$ and $B = \{b_1, b_2, b_3\}$ with group A making the offers, that satisfies the following property:

- One of the B 's can lie—during the execution of the Gale-Shapley algorithm she may decide to accept or reject an offer differently from what the algorithm tells her to do based on her true preferences. She can lie only once, but because of her lie, the solution will yield a stable matching, where at least one of the B 's gets a partner that she prefers to the one she gets in the honest execution, and none of the B 's gets a worse partner.

You should specify the preference lists, simulate the honest execution, point out where a lie should occur, and simulate the execution with the lie, verifying that the outcome is as claimed above.

One possible solution:

$a_1: b_1, b_2, b_3$
 $a_2: b_1, b_3, b_2$
 $a_3: b_2, b_1, b_3$

b_1 : a_3, a_1, a_2

b_2 : a_1, a_3, a_2

b_3 : a_2, a_1, a_3

Honest execution:

1. a_1 makes offer to b_1 : accepted as b_1 is unmatched
Current partial matching: (a_1, b_1)
2. a_2 makes offer to b_1 : rejected as b_1 prefers a_1 to a_2 $\leftarrow$ lie should occur here
Current partial matching: (a_1, b_1)
3. a_3 makes offer to b_2 : accepted as b_2 is unmatched
Current partial matching: $(a_1, b_1), (a_3, b_2)$
4. a_2 makes offer to b_3 : accepted as b_3 is unmatched
Final matching: $(a_1, b_1), (a_3, b_2), (a_2, b_3)$

Execution with lie:

1. a_1 makes offer to b_1 : accepted as b_1 is unmatched
Current partial matching: (a_1, b_1)
2. a_2 makes offer to b_1 : (lie) accepted although b_1 prefers a_1 to a_2
Current partial matching: (a_2, b_1)
3. a_3 makes offer to b_2 : accepted as b_2 is unmatched
Current partial matching: $(a_2, b_1), (a_3, b_2)$
4. a_1 makes offer to b_2 : accepted as b_2 prefers a_1 to a_3
Current partial matching: $(a_2, b_1), (a_1, b_2)$
5. a_3 makes offer to b_1 : accepted as b_1 prefers a_3 to a_2
Current partial matching: $(a_1, b_2), (a_3, b_1)$
6. a_2 makes offer to b_3 : accepted as b_3 is unmatched
Final matching: $(a_1, b_2), (a_3, b_1), (a_2, b_3)$

b_1 and b_2 get partners they prefer and b_3 gets the same partner.

Hints

1c. Use parts a and b and the properties of the Gale-Shapley algorithm to determine the set of possible partners for each person in a stable matching and then work from there.

2b. Taking the logarithm of these functions may be helpful for comparing them.

3b. Can you make the upper bound on the total number of offers be less than n^2 ?